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The third criterion of congruence of triangles

Three sides and nothing else — the criterion that makes a triangle rigid.

LESSON 53–54 | 2 × 40 MIN | MATEMATIKA 7, §23 | S8 5

LESSON CLOCK

15' | 25' | 25' | 20' | 5'

The criterion	15 min
Rigidity	25 min
Using it	25 min
Choosing a criterion	20 min
Homework	5 min

LEARNING OBJECTIVES

- State and apply the SSS criterion.
- Use it to prove properties of the rhombus and the kite.
- Explain why a triangle is rigid and a quadrilateral is not.
- Choose between the three criteria from the given data.

TEXTBOOK

National	pp. 143–148
Cambridge	Stage 8, pp. 48–56
Workbook	Exercise 5.1

01 Explanation

The criterion

> If the three sides of one triangle are equal to the three sides of another, the triangles are congruent

$$AB = A_1B_1,\ BC = B_1C_1,\ CA = C_1A_1 \Longrightarrow \triangle ABC \equiv \triangle A_1B_1C_1$$

No angle is mentioned at all, and yet all three angles turn out equal. Three lengths determine a triangle completely.

THIS IS THE CRITERION A COMPUTER WOULD USE

It needs no reasoning about the figure — only three comparisons. Every question that supplies six lengths and no angles is answered by it.

Rigidity

A framework of three rods joined at their ends cannot change shape; one of four rods can.

FRAMEWORK	RIGID?	WHY
triangle	yes	three sides fix it — the third criterion
quadrilateral	no	it can be pushed into a parallelogram
quadrilateral with one diagonal	yes	it becomes two triangles

This is why bridges, roof trusses and pylons are built from triangles, and why a garden gate is braced with a diagonal.

THE THIRD CRITERION IS AN ENGINEERING FACT

Every triangulated structure in the world depends on it. The theorem and the brace on a gate are the same statement.

Using it

Problem. In a rhombus $ABCD$, prove that the diagonal AC bisects the angles at A and C.

STATEMENT	REASON
$AB = AD$	a rhombus has four equal sides
$CB = CD$	the same
$AC = AC$	common
$\triangle ABC \equiv \triangle ADC$	third criterion
$\angle BAC = \angle DAC$	CPCT

THE COMMON SIDE AGAIN

As in the first criterion, the shared side supplies the third equality without being given. Look for it first in every figure.

Choosing a criterion

GIVEN	CRITERION
two sides and the angle between them	the first (SAS)
a side and two angles	the second (ASA)
three sides	the third (SSS)
two sides and a non-included angle	none — not enough
three angles	none — similarity only

NAME THE CRITERION IN EVERY PROOF

A proof that reaches the right conclusion without naming which criterion was used is incomplete, and is marked as such in both the national and the Cambridge papers.

02 Worked examples

EXAMPLE 1 **Two triangles have sides $5, 6, 7$ and $7, 5, 6$. Are they congruent?**

1. Sort both lists: $5, 6, 7$ and $5, 6, 7$.
2. All three pairs are equal.
3. By the third criterion, congruent.
4. The order in which they were written does not matter.

ANSWER

Yes, by SSS

EXAMPLE 2 **In rhombus *ABCD*, prove $\triangle ABC \equiv \triangle ADC$.**

1. $AB = AD$ — all sides of a rhombus are equal.
2. $CB = CD$ — the same.
3. AC is common.
4. Third criterion.

ANSWER

Proved by SSS

EXAMPLE 3 **Why is a four-rod framework not rigid, and how is it braced?**

1. Four sides do not determine the angles.
2. The frame can be pushed out of square.
3. Adding one diagonal makes two triangles.
4. Each is rigid by SSS, so the whole frame is.

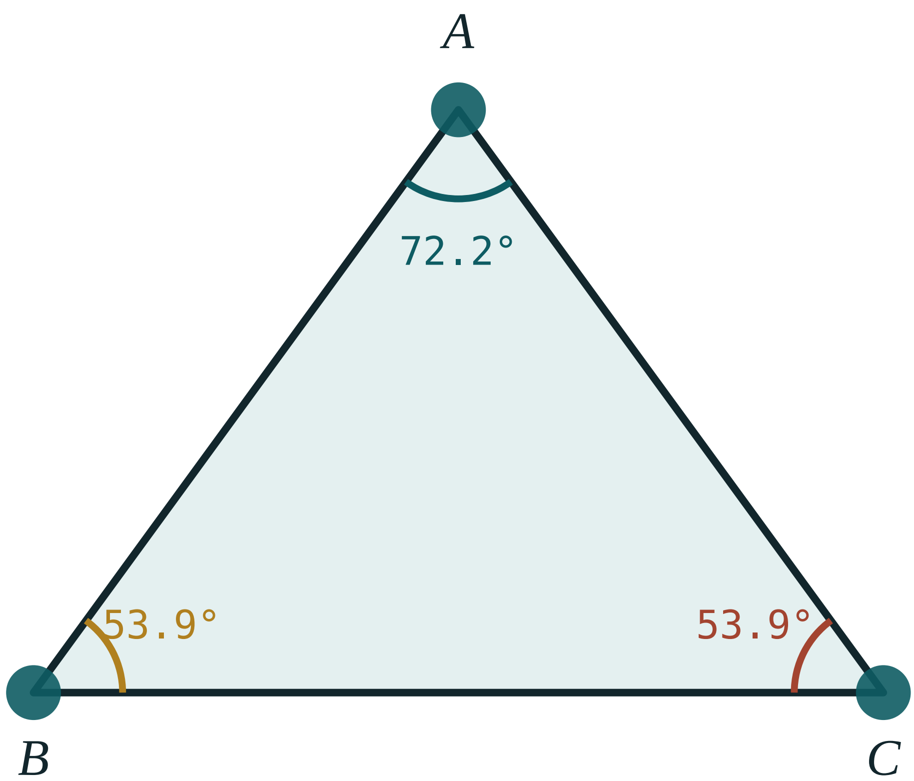
ANSWER

Add a diagonal

03 Interactive model

Pin three strips of card into a triangle and four into a quadrilateral; the class pushes both, and the third criterion becomes visible in ten seconds.

Three sides fix the triangle



∠A **72.2°** | ∠B **53.9°** | ∠C **53.9°** | sum **180°** | longest side **BC** | largest angle **∠A**

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Third criterion	Uchinchi alomat	Третий признак
Three sides	Uchta tomon	Три стороны
Rigid	Qattiq	Жёсткий
Rhombus	Romb	Ромб
Kite	Deltoid	Дельтоид
Diagonal	Diagonal	Диагональ
Framework	Karkas	Каркас
To choose	Tanlash	Выбрать

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

The third criterion needs:

- three angles
- three sides
- two sides and an angle
- a side and two angles

A triangular framework is:

- flexible
- rigid
- unstable
- impossible

A quadrilateral framework is made rigid by:

- a fifth rod along a side
- a diagonal
- nothing
- a shorter rod

Sides $5, 6, 7$ and $7, 5, 6$ give:

- congruence
- similarity only
- nothing
- an error

Two sides and a non-included angle give:

- congruence
- not necessarily
- similarity
- nothing at all

A complete proof must:

- give the answer
- name the criterion
- include measurements
- be short

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. The third criterion in letters

   ANSWER *SSS*

2. Sides $5, 6, 7$ and $5, 6, 7$

   ANSWER Congruent

3. Sides $5, 6, 7$ and $5, 6, 8$

   ANSWER Not congruent

4. Is a triangle rigid?

   ANSWER Yes

5. Is a quadrilateral rigid?

   ANSWER No

6. How is it braced?

   ANSWER With a diagonal

7. A shared side gives

   ANSWER An equality for free

Medium · the standard question

7 PROBLEMS

1. Sides $5, 6, 7$ and $7, 5, 6$

   ANSWER Congruent

2. Which criterion for six lengths?

   ANSWER The third

3. Which for a side and two angles?

   ANSWER The second

4. Which for two sides and the included angle?

   ANSWER The first

5. Which for three angles?

   ANSWER None

6. In a rhombus, $\triangle ABC$ and $\triangle ADC$

   ANSWER Congruent by SSS

7. What follows about $\angle BAC$ and $\angle DAC$?

   ANSWER They are equal

Hard · stretch and reason

7 PROBLEMS

1. Prove that a diagonal of a rhombus bisects its angles

   ANSWER SSS, then CPCT

2. Prove that the diagonals of a rhombus are perpendicular

   ANSWER Congruent triangles at the centre

3. Prove that a kite has one diagonal as an axis of symmetry

   ANSWER SSS on the two triangles

4. Why is a bridge built of triangles?

   ANSWER Only triangles are rigid

5. Two triangles with sides a, b, c and c, b, a

   ANSWER Congruent

6. Can two triangles have five equal elements and not be congruent?

   ANSWER Yes — similar triangles can

7. How many rods brace a quadrilateral?

   ANSWER One diagonal

07 Homework

Homework — 5 tasks

Sort both lists of sides before comparing, and name the criterion in every proof.

1. Are triangles with sides $8, 9, 10$ and $10, 8, 9$ congruent?
2. Prove that a diagonal of a rhombus bisects two of its angles.
3. Explain why a triangle is rigid and a quadrilateral is not.
4. State which criterion applies to each of the five cases in the table.
5. Prove that the diagonals of a rhombus are perpendicular.